



**Arab republic of Egypt
Ministry of health & population
General Administration of Pharmaceutical
Affairs**

Albumin Restriction Protocol

الإصدار الاول لجنة السياسات الدوائية

April-2025



PREFACE

The Egyptian technical operational manuals and guides are published under the authority of the General Administration for Pharmaceutical Affairs (GAPA), Ministry of Health and Population (MOHP) within the framework of the strategic plan to raise the quality of the practice of pharmaceutical care and the rational use of medications, including antimicrobials, for patient benefit.

*GAPA, the entrusted authority for organizing, supervising, and following up with the workflow in collaboration with MOHP sectors, directorates in the pharmaceutical sector, and hospitals of the MOHP issue series of pharmaceutical papers that focus on Medication Management and pharmacy practice standards, including which describes the workflow of pharmacy service and explanation of the necessary documentation involved to raise the quality of pharmaceutical care services provided to the patient and professional competence of pharmacists at health care settings . New and Important issues shall be published soon regarding operational manuals and job descriptions referring to pharmaceutical care regulations and pharmacist job descriptions which are able to steer excellent management practice in conducive environments toward fulfillment of customers' needs. This Issue No.8, **Albumin Use protocol** includes a guidance for setting restrictive interventions on prescribing of albumin to minimize its misuse and subsequent wastage*

Acknowledgment

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List of abbreviations

BUN: Blood urea nitrogen

HAS: Human Albumin Solution

SCr: serum creatinine

T. bil.: Total Bilirubin

Scope of the Protocol: This protocol is aimed to guide and restrict the use of human albumin solution in adult patients.

Albumin Restriction Protocol

Indications, dosage and administration for human Albumin:

General Recommendations

“The concentration of the albumin preparation, dosage, and the infusion rate should be adjusted according to the indication and the patient’s requirements.”

The dose required depends on the weight of the patient, the severity of trauma or illness and on continuing fluid and protein losses. Dosing depends on the measures of adequacy of circulating volume and not on plasma albumin levels only.

N.B.: Albumin is not indicated to be given in hypoalbuminemia alone

The daily dose should not exceed 2 g of Albumin (Human) per kg of body weight.

Indication	Dosage	Notes
<u>Prevention of Central Volume Depletion after Paracentesis due to Cirrhotic Ascites (Treatment Adjunct)</u> <u>Can use 20% HAS</u>	Large volume paracentesis: More than 5 liters removed IV: 6 to 8 g for every liter removed or 50gm total for paracentesis more than 5 liter	Note: Administer soon after the procedure to avoid postprocedural complications (ex. hypovolemia, hyponatremia, renal impairment)
<u>Spontaneous bacterial peritonitis, treatment:</u> <u>Can use 20% HAS</u>	IV: Initial: 1.5 g/kg within 6 hours of diagnosis in combination with appropriate antimicrobial therapy, followed by 1 g/kg on day 3	for patients with creatinine >1 mg/dL, BUN >30 mg/dL, or bilirubin >4 mg/dL
<u>Plasma exchange, therapeutic:</u> <u>Use 5% HAS</u>	IV: Titrate dose to plasma volume removed during procedure and serum fibrinogen levels.	The use of albumin is appropriate only for the exchange of some of large volumes of plasma: more than 20 mL/kg in a single session or 20mL/kg/week in successive sessions. -To avoid excessive hypoproteinemia, be cautious during certain types of exchange



		transfusion, especially when using very large volumes or previously frozen or washed red blood cells.
<u>Hepatorenal Syndrome - Acute Kidney Injury (Hepatorenal Syndrome Type 1), Diagnosis and Treatment (Adjunctive Agent)</u> <u>Use 20% HAS</u>	For diagnosis: Albumin 20% 1 g/kg/day for 2 days (dose up to a maximum of 100 g per day) For Treatment Albumin 20% 20-50 g daily	-Used in cirrhotic patients with Acute kidney injury stage more than 1A. -Human albumin is used before or in combination with norepinephrine, terlipressin, or midodrine plus octreotide. -Treatment should be maintained until a complete response (Serum creatinine within 0.3 md/dl from baseline value) or partial response (regression of AKI stage to a final serum creatinine $\geq$ 0.3mg/dl from baseline value) or a maximum of 14 days -Discontinue vasoconstrictor and albumin therapy in patients with HRS-AKI who meet any of the following criteria during treatment: - No improvement in serum creatinine after three days of therapy - Serum creatinine improved to within 0.3 mg/dL of baseline (complete response) - Serious adverse reaction - Initiation of kidney replacement therapy - Liver transplantation - Total duration of therapy 14 days
<u>Post -operative liver Transplantation</u> <u>Use 20% HAS</u>	25 grams per day until the serum albumin level reaches 2.5 grams/dL. If the serum albumin remains	Albumin may be useful for postoperative liver transplant patients in the control of ascites and



	below 2.5 grams/dL, continue albumin dosing for up to 4 days, and consult liver transplant surgeons afterwards to consider further use.	peripheral edema if all of the conditions are met: 1. Serum albumin is < 2.5 g/dl; 2. Pulmonary capillary wedge pressure is < 12 mm Hg; 3. Hematocrit is > 30%
<u>Post Liver resection</u> <u>Use 20% HAS</u>	25 gm/day until albumin is ≥2.5 gm/dL, continue albumin dosing for up to 4 days and consult liver surgeons for consideration of ongoing use	Albumin is used in patients where the liver resected more than 40% and with serum albumin < 2.5 g/dL if crystalloids alone fail to achieve adequate intravascular volume and unavailability of non-protein colloid solutions or the presence of contraindications of non-protein colloid use.
<u>Special Considerations</u>		
<u>Burns (occasionally appropriate indication)</u> <u>Use 5% HAS</u>	In the post-resuscitation phase, once the problems of circulatory volume caused by the marked capillary permeability have been overcome. (indicated after 24 hr. post burn if severe albumin depletion in order to favor edema reabsorption) Amount of albumin required to achieve adequate plasma volume and protein content should be determined by direct observation of vital signs or measurement of either plasma oncotic pressure or protein content	Albumin is used as an alternative in cases of unavailability of non-protein colloid solutions (if crystalloids fail to achieve adequate intravascular volume) or Presence of contraindications of non-protein colloid use.



Procedure for Albumin Restriction Form:

- This form is used as a guide for physicians and clinical pharmacist to inform on the use of albumin and limit its wastage.
- The physician fills patient's demographic data and the specific indication for the use of albumin with the dosage and method of administration
- The clinical pharmacist then fills their own recommendation (acceptance/rejection) and signs the form
- The clinical pharmacist is then required to perform daily follow up to ensure the continuous use of albumin is done for correct indication.
- N.B.: Please refer to the physician's opinion in case of Albumin shortage



Albumin Restricted Use Form

Date: -----

Patient information:			
Patient Name:	File No.:	Department:	Ward:
Age:	Gender:	Allergies:	Weight:

Indications:		
☐ Cirrhotic ascites ☐ Spontaneous Bacterial Peritonitis ☐ Hepatorenal Syndrome ☐ Plasmapheresis ☐ Organ Transplant ☐ Liver resection ☐ Hypovolemia ☐ Burns	Albumin Level:	Blood urea nitrogen:
	Volume of liters removed:	Total Bilirubin:
	Serum Creatinine:	Poly-morph neutrophils in ascitic fluid:

Dosage Regimen	
Dose:	Duration:
Frequency:	Infusion Rate:

Filled by Prescribing Physician	
Name:	Signature:

Filled by Clinical Pharmacist	
Recommendation:	
Name:	Signature:

Follow up:(filled by clinical pharmacist)		
Day 1:	Alb. Level:	Rec:
Day 2:	Alb. Level:	Rec:
Day 3:	Alb. Level:	Rec:
Day 4:	Alb. Level:	Rec:



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